NAME: _____ Geometry (FL B.E.S.T.)

7.2 Properties of Parallelograms

A parallelogram is a quadrilateral with both pairs of opposite sides parallel. From that one fact, four powerful properties follow: opposite sides are congruent, opposite angles are congruent, consecutive angles are supplementary (180° total), and the diagonals bisect each other. These properties come from parallel lines cut by a transversal (each side of a parallelogram acts as a transversal across the opposite pair of parallel sides) and from congruent triangles formed by a diagonal. This worksheet uses all four properties together — often two variables in one problem — and asks you to justify, not just compute.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Parallelogram: $AB = 15$, $AD = 9$. Find CD and BC.

2. $\angle A = 125°$. Find $\angle B$ and $\angle C$.

3. Diagonals meet at O; $BD = 18$. Find BO.

4. $\angle A = 2x + 10$, $\angle C = 3x - 20$. Find x.

5. $\angle A = 4x$, $\angle B = 5x + 9$. Find x.

6. $AB = 4x + 1$, $CD = 3x + 6$. Find AB.

7. $AO = x + 3$, $OC = 2x - 1$. Find AC.

8. $\angle D = 70°$. Find $\angle A$.

9. Do diagonals of a parallelogram always bisect each other?

10. $AD = 7x - 4$, $BC = 5x + 8$. Find AD.

11. $\angle A = 100°$. Find consecutive angle $\angle D$.

12. $AO = 6$. Find AC.

13. $\angle B = 3x + 15$, $\angle C = 2x + 25$. Find $\angle B$ (note $\angle B$, $\angle C$ are consecutive).

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14. In parallelogram $ABCD$, $m\angle A = (5x - 15)^\circ$ and $m\angle B = (3x + 35)^\circ$. Diagonals meet at O with $AO = 2y + 4$ and $OC = 5y - 11$. Find x, y, $m\angle A$, and AC.

15. A parallelogram has perimeter 100, and one side is 8 more than twice the other. Find all four side lengths.

✓ Answer Key

1. $CD = 15$; $BC = 9$
2. $\angle B = 55^\circ$; $\angle C = 125^\circ$
3. 9
4. $x = 30$
5. $x = 19$
6. 21 ($x = 5$)
7. 14 ($x = 4$)
8. 110°
9. Yes
10. 38 ($x = 6$)
11. 80°
12. 12
13. $\angle B = 111^\circ$ ($x = 32$)
14. $x = 20, y = 5$; $m\angle A = 85^\circ$; $AC = 28$
15. 14, 36, 14, 36